DRAFT — P1–P5 MEASURED; P4-a-block-energy HIT

A3 Practice: Counterfactual Prediction Quiz + Answer Key

CS 312 · Optimization 2 · **Hand-in PDF #2** (of 2) · Practice only (not the official Oct 20 quiz)
Modeled on A2 *Outcome Prediction Quiz: Hyperparameter Scaling* (`internal/hyperparameter_scaling_prediction_quiz.txt`).

PDF role: practice quiz + answer key. Results / intuition / quantitative properties → `a3-results-intuition.md` / `media/pdfs/a3-results-intuition.pdf`. See `a3-deliverables-readme.md`.

Ledger alignment: every item cites a stable ledger ID (`docs/a3-prediction-ledger.md` / `.csv`). Kind tags there: `predicted` · `measured` · `fitted` · `extrapolated`.

Instructions. Predict outcomes of the following experiments. Each question has two parts:

- **From the assignment** restates the assignment setting (reference only).
- **This question changes** lists every difference; everything else is unchanged.

Check one box per row. Scoring (when used for self-study): correct option = full credit; **adjacent interval = half**. Each question has three rationale fields: assignment problem cited, what that experiment showed (with numbers — fill only after measurements), and why it implies your answer.

Default setup. Course d8 LM: ~35M params, ~614M tokens unless stated, AdamW (\beta_1,\beta_2=(0.9,0.95)), (\epsilon=10^{-8}), context 1024, batch 64, WD 0.1 (masked), grad clip 1.0; lower validation loss is better. Problem-specific overrides (e.g. P1 `wsd0.0`, P3 full-batch GD) are stated in each item.

Conventions.

- Barrier (B(\theta_1,\theta_2)=L((\theta_1+\theta_2)/2)-(L(\theta_1)+L(\theta_2))/2); averaging works if (B\le 0.01).
- Gap (G=L(\theta_{\mathrm{avg}})-L(p_{\mathrm{ens}})); ensemble (p_{\mathrm{ens}}=\tfrac12 p_{\theta_1}+\tfrac12 p_{\theta_2}).
- Regret (R_M(t)=L(\theta^M_t)-L(\theta^{\mathrm{ref}}_t)).
- GD edge: (\eta\lambda_{\max}=2). Adam (\beta_1=0.9) frozen-precond story: (\eta\lambda^{\mathrm{pre}}_{\max}=38).

Evidence labels (answer key).

Label	Meaning
PREDICTION	Pre-registered before the corresponding run / fit; mechanism + handout thresholds. Not ground truth.
MEASURED	Direct experimental cell from Batch 1 (`media/p1/*`) or Batch 2 finals (`media/p2/batch2_partial_finals.csv`).
FITTED	Value from Appendix Q1.1 Fits A–C on MEASURED points (in-sample).
EXTRAPOLATED	Out-of-grid interval from those fits (ledger Kind=`extrapolated`); not a new measurement.

Do **not** invent MEASURED numbers. Leave Resolved blank / **PENDING** until the required final metric exists.

Batch 2 lock (P2/P3). Q5–Q8, Q11–Q12, and Q22–Q24 PREDICTION slots were locked at `2026-10-11T00:32:07Z` **before Batch 2 completion**. Mid-run W&B was **not** used to tune bands. At **9/9**: Q6a dip **HIT**; Q11a/Q23 HF (\Delta L) **MISS** + vs-invsqrt **HIT**; Q22 schedule resolved; Q12a/Q24a (\lambda_{\max}=245.16314737956807) **HIT**; Q8a/Q24b (\eta\lambda^{\mathrm{pre}}=30.987715914514) **HIT**. Still blank/PENDING: Q6b mid-dip, Q5 EMA. Ledger: `docs/a3-prediction-ledger.md` / `.csv`.

P4 Batch 3 lock. Q9–Q10 and Q26–Q29 PREDICTION bands are locked from `docs/a3-p4-preregister-draft.md` **before** Batch 3 measure/train. Stage a **13/13** MEASURED ((\lambda_{\max}@9375=5.967310256347902)). Ledger apply (resolvable IDs): `P4-a-family-sharpest` **MISS** (late=`head`); `P4-a-norm-sharpness` **HIT**; `P4-a-density-neg` **MISS** (no shrink 0→9375); `P4-lam-9375-earlyfit` **MISS** (above `[0.3,4.0)`); `P4-lam-1526-earlyfit` **HIT** (prior). `P4-a-block-energy` **HIT** (excess_final=0.08727119221832025; fam_above=True; layer_above=True) — Q27b Resolved filled. Q9: `P4-invar-dL` **MISS** (|ΔL|=2.73853462934494; ratio=0.4658784038724411). Q10a **MISS** (mean=0.09296940652964016); Q10b **MISS** (gap=-0.0003446936607360 84). Q28–Q29 PENDING (optional arms untouched).

P5 exploratory lock. MEASURED from `internal/p5-score-dryrun.md` / `media/p5/`: `P5-B-cool-x0` **MISS**, `P5-dB-cool` **MISS**, `P5-B-const-ctrl` **HIT**, `P5-rank-arms` **MISS**, `P5-G-cool-fitC` **HIT**, `P5-emergent-basin` **MISS**; `P5-L-cool-dir` **INSUFFICIENT** (`dL=null` / (L_R,L_C) missing — no HIT/MISS). PREDICTION key retained **(a)1 · (b)4 · (c)1**. Keep locked bands; never erase a miss.

Prediction

Q1. Mode connectivity: early vs late fork

Difficulty: Easy · theme P1(b) · ledger `P1-B-early-long`

From the assignment (reference only). P1 forked-run grid with `wsd0.0` (constant after warmup), shared prefix (x), branch length (y), different `batch_order_seed`, barrier criterion (B\le 0.01).

This question changes. Compare only two cells: (A=(x,y)=(0,500)) and (C=(1000,25)). Same peak LR, same seeds `{1,2}`.

(a) Barrier at cell (A=(0,500)) (10 points). Which interval contains (B(0,500))?

· 1. ((-\infty, 0.01]) · 2. ((0.01, 0.05)) · 3. ([0.05, 0.2)) · 4. ([0.2, \infty))

(b) Barrier at cell (C=(1000,25)) (10 points). Which interval contains (B(1000,25))?

· 1. ((-\infty, 0.01]) · 2. ((0.01, 0.05)) · 3. ([0.05, 0.2)) · 4. ([0.2, \infty))

Mark answers and rationale on the answer sheet.

Q2. Ensemble vs weight average when (B) is large

Difficulty: Medium · theme P1(a–b) · ledger `P1-ens-gap`

From the assignment. For each grid cell you evaluate endpoints, (0.5/0.5) weight average, and (0.5/0.5) prediction ensemble (p_{\mathrm{ens}}=\tfrac12 p_{\theta_1}+\tfrac12 p_{\theta_2}).

This question changes. Restrict to cells with (B>0.01). Let (G=L(\theta_{\mathrm{avg}})-L(p_{\mathrm{ens}})).

(a) Sign / ranking (8 points). Which do you predict?

· 1. (G<0) (weight avg better) · 2. (G\in[0,0.01)) · 3. (G\ge 0.01) and ensemble better · 4. Ensemble and weight avg within (0.005) always

(b) Magnitude (12 points). Which interval contains the median (G) over those cells?

· 1. ([0, 0.02)) · 2. ([0.02, 0.08)) · 3. ([0.08, 0.2)) · 4. ([0.2, \infty))

Q3. Interpolation peak vs midpoint barrier

Difficulty: Medium · theme P1(c) · ledger `P1-interp-peak`

From the assignment. Full path (\theta(\alpha)=(1-\alpha)\theta_1+\alpha\theta_2), (\alpha\in{0,0.25,0.5,0.75,1}).

This question changes. Take one representative cell with midpoint (B\in[0.05,0.2)). Let (L_\alpha=L(\theta(\alpha))).

(a) Ranking (10 points). Which ranking is closest?

· 1. (L_{0.5}) strictly largest among the five (\alpha)
· 2. (L_{0.25}) or (L_{0.75}) largest (asymmetric barrier)
· 3. Path nearly flat: (\max L_\alpha-\min L_\alpha<0.01)
· 4. Endpoints worst; midpoint best (inverted)

(b) Barrier height check (10 points). Which interval contains (L_{0.5}-\frac{L_0+L_1}{2})?

· 1. ([0, 0.01)) · 2. ([0.01, 0.05)) · 3. ([0.05, 0.2)) · 4. ([0.2, \infty))

Q4. Schedule decay lowers barrier at fixed fork

Difficulty: Hard · theme P1(d)+P2 · ledger `P1-d-dB-cos`
(Evidence: PREDICTION only — no cosine Batch 2 cells yet.)

From the assignment. P1(d) asks which training choices control same-basin membership via (B) and interpolation curves.

This question changes. Fix ((x,y)=(250,250)). Arm W uses `wsd0.0` after fork; arm C uses cosine decay after fork (same peak LR, same seeds). Define (\Delta B=B_W-B_C).

(a) (10 points). Which interval contains (\Delta B)?

· 1. ((-\infty, 0)) · 2. ([0, 0.02)) · 3. ([0.02, 0.1)) · 4. ([0.1, \infty))

(b) (10 points). After the change, does arm C satisfy (B_C\le 0.01)?

· 1. Yes · 2. No · 3. Depends on (B_W) band · 4. Always yes regardless of (B_W)

(Study note: MEASURED (B_W(250,250)=-0.0779\le 0.01) already under `wsd0.0`*; cosine (\Delta B) is still unmeasured.)*

Q5. EMA closes schedule gaps

Difficulty: Medium · theme P2(b) · ledger `P2-ema-gap-frac`
(Evidence: PREDICTION — Batch 2 locked `2026-10-11T00:32:07Z`**; jobs 1–4 schedule+EMA. No finals used.)**

From the assignment. Compare `wsd0.0`, `linear`, `cos`, `wsd0.2` at equal tokens; EMA decays ({0.99,0.999}) eval-only; regret vs best raw final.

This question changes. Let (G_{\mathrm{raw}}=L^{\mathrm{raw}}*{\mathrm{worst}}-L^{\mathrm{raw}}*{\mathrm{best}}) among the four schedules (best peak LR each). Let (G_{\mathrm{EMA}}) be the same gap using each run's best EMA val. Define closed fraction (f=1-G_{\mathrm{EMA}}/G_{\mathrm{raw}}) (if (G_{\mathrm{raw}}=0), treat (f=1)).

(a) (10 points). Which interval contains (f)?

· 1. ([0, 0.25)) · 2. ([0.25, 0.5)) · 3. ([0.5, 0.75)) · 4. ([0.75, 1])

(b) (10 points). After best EMA, which schedule still has the lowest val loss?

· 1. `wsd0.0` · 2. `linear` · 3. `cos` or `wsd0.2` · 4. All four within (0.01)

Q6. LR dip then rewarm — final regret

Difficulty: Easy · theme P2(c) · ledger `B2-P2-dip-R` / `P2-dip-final-R`
(Evidence: MEASURED final (R) — job4 `rms7iyyt` **vs job3** `wsd0.2` **control; mid-dip unscored.)**

From the assignment. Temporary `lr_dip` then rewarm without resetting state; control shares final cooldown; plot regret vs control.

This question changes. One dip depth/duration (moderate: scale (0.1), duration (10%) of training, mid-run). Let final (R=L_{\mathrm{dip}}-L_{\mathrm{ctrl}}).

(a) Final regret (12 points). Which interval contains (R)?

· 1. ((-\infty, -0.02)) · 2. ([-0.02, 0.02)) · 3. ([0.02, 0.08)) · 4. ([0.08, \infty))

(b) During-dip regret (8 points). Mid-dip (R_{\mathrm{mid}}) (at bottom of dip) is in:

· 1. ((-\infty, -0.02)) · 2. ([-0.02, 0.02)) · 3. ([0.02, 0.08)) · 4. ([0.08, \infty))

Q7. Edge of stability: mid-run (\eta) changes

Difficulty: Easy · theme P3(a)ii · ledger `P3-eta-half` / `P3-eta-double`
(Evidence: PREDICTION / PENDING — η-change arms not in Batch 2 required 9; clean-edge P3 is MEASURED elsewhere as Q12/Q24a.)

From the assignment. Full-batch GD, (N=32), (\eta=0.01), 1500 updates; progressive sharpening toward (2/\eta=200). At update 750: either `step750x0.5` or double (\eta).

This question changes. Report median (\lambda_{\max}) over updates (1200)–(1500).

(a) After (\eta\leftarrow 0.005) (10 points). Interval for median (\lambda_{\max}):

· 1. ([150, 250)) · 2. ([250, 350)) · 3. ([350, 450)) · 4. ([450, \infty))

(b) After (\eta\leftarrow 0.02) (10 points). Interval for median (\lambda_{\max}):

· 1. ((0, 80)) · 2. ([80, 120)) · 3. ([120, 180)) · 4. ([180, \infty))

Q8. Minibatch AdamW preconditioned edge

Difficulty: Medium · theme P3(b) · ledger `B2-P3-adamw-pre` / `P3-mb-edge`
(Evidence: MEASURED — job `a3-b2-p3-1-adamw` / `foew12y0`**; post-warmup median (\eta\lambda^{\mathrm{pre}}=30.987715914514) from** `media/p2/p3_adamw_sharpness_summary.json`**.)**

From the assignment. `minibatch()` recipe; plot (\eta_t\lambda^{\mathrm{pre}}_{\max}) after warmup; Adam (\beta_1=0.9) frozen-precond threshold (38).

This question changes. Summarize the post-warmup median of (\eta_t\lambda^{\mathrm{pre}}_{\max}) on the held-out probe.

(a) (12 points). Which interval contains that median?

· 1. ([0, 20)) · 2. ([20, 30)) · 3. ([30, 50)) · 4. ([50, \infty))

(b) (8 points). Relative to full-batch GD's (\eta\lambda\approx 2) ride, minibatch AdamW is:

· 1. Exactly as tight around its threshold · 2. Noisier / wider band around ~38 · 3. Stuck near 2 (not 38) · 4. Diverges ((\eta\lambda^{\mathrm{pre}}\to\infty))

Q9. Loss-invariant rescale moves sharpness

Difficulty: Hard · theme P4(b) · ledger `P4-invar-dL`
(Evidence: MEASURED — `P4-invar-dL` **MISS; |ΔL|=2.73853462934494; λ ratio=0.4658784038724411; λ 5.967310256347902→2.780040977639008; probe_loss 2.9474442899227142→5.685978919267654)

From the assignment. Default d8 AdamW finals: (\lambda_{\max}\in[4.5,6.0]), val (\in[2.925,2.931]) (handout bands). Find transforms / recipes that move (\lambda_{\max}) by (\ge 2\times) without leaving the loss band.

This question changes. Apply a hand loss-invariant rescale to one weight family on the final checkpoint (no further training; stage `b-rescale`).

(a) (10 points). (\lvert\Delta L_{\mathrm{val}}\rvert) is in:

· 1. ([0, 0.002)) · 2. ([0.002, 0.01)) · 3. ([0.01, 0.05)) · 4. ([0.05, \infty))

(b) (10 points). Ratio (\lambda_{\max}^{\mathrm{after}}/\lambda_{\max}^{\mathrm{before}}) is in:

· 1. ((0, 0.5)) · 2. ([0.5, 0.9)) · 3. ([0.9, 1.1)) · 4. ([2, \infty))
(If your rescale direction lowers sharpness, treat option 1 as the success band for (\ge 2\times) decrease; option 4 for (\ge 2\times) increase.)

Q10. Top-10 subspace continue (SGDM)

Difficulty: Medium · theme P4(c) · ledger `P4-top10-frac` / `P4-c-loss-rank`
(Evidence: MEASURED — `P4-top10-frac` / `P4-c-loss-rank` from `internal/p4-c-score-dryrun.md`; mean=0.09296940652964016 **MISS; gap=-0.0003446936607360 84 **MISS**)

From the assignment. From SGDM checkpoint at 100M tokens (step 1526): zero momentum; continue 300 steps with full / top-10 / removed updates; refresh eigenvectors every 25 steps (`k=10`).

This question changes. Report mean gradient fraction in the top-10 span over the continue, and final loss ranking.

(a) (10 points). Mean (|P_{10}g|/|g|) in:

· 1. ([0, 0.3)) · 2. ([0.3, 0.6)) · 3. ([0.6, 0.85)) · 4. ([0.85, 1])

(b) (10 points). Final val-loss ranking (lowest → highest):

· 1. full (\approx) top-10 (<) removed
· 2. full (<) removed (<) top-10
· 3. removed (<) full (\approx) top-10
· 4. all three within (0.01)

Q11. Horizon-free vs budget-tuned cooldown

Difficulty: Hard · theme P2(d) · ledger `P2-horizon-dL` / `B2-P2-invsqrt-dL` / `B2-P2-sgdr-dL`
(Evidence: MEASURED invsqrt + SGDR (\Delta L) vs linear @ Batch 2 9/9 — both band MISS; mid-budget stop still PREDICTION / PENDING.)

From the assignment. Invsqrt (\alpha_t=\alpha_{\max}\min{t/w,\sqrt{w/t}}) and SGDR cosine warm restarts; (w,T,\alpha) bounds chosen independently of final budget; no budget-specific cooldown; AdamW unchanged; compare val and regret to schedules tuned to each budget.

This question changes. Fix one invsqrt ((w,\alpha_{\max})) and one SGDR ((T,\alpha_{\max}, \alpha_{\min})) pair. Evaluate at two stopping budgets: mid (T_{\mathrm{mid}}) and long (T_{\mathrm{long}}) where the budget-tuned baseline gets a long final cooldown. Let (\Delta L = L_{\mathrm{horizon\text{-}free}}-L_{\mathrm{tuned}}) at each stop (best of invsqrt/SGDR vs best tuned cosine/`wsd0.2`).

(a) At (T_{\mathrm{long}}) (10 points). Which interval contains (\Delta L)?

· 1. ((-\infty, -0.01)) · 2. ([-0.01, 0.01)) · 3. ([0.01, 0.08)) · 4. ([0.08, \infty))

(b) At (T_{\mathrm{mid}}) (10 points). Ranking of usefulness:

· 1. Horizon-free clearly better ((\Delta L<-0.01))
· 2. Competitive: (\lvert\Delta L\rvert<0.02)
· 3. Horizon-free clearly worse ((\Delta L\ge 0.08))
· 4. Diverges / NaN

Q12. Full-batch clean edge ((N=32), (\eta=0.01))

Difficulty: Easy · theme P3(a)i · ledger `B2-P3-sgd-lam` / `P3-clean-edge`
(Evidence: MEASURED — job `a3-b2-p3-0-sgd` / `0dewkykj`**; median (\lambda_{\max} =245.16314737956807) from** `media/p2/p3_sgd_sharpness_summary.json`**.)**

From the assignment. `full_batch(32, 1500, learning_rate=0.01)`; GD; 16× embedding init; plot (\lambda_{\max}) vs update and vs train loss; dashed threshold at (2/\eta).

This question changes. Report the median (\lambda_{\max}) over updates (1200)–(1500) on the clean case (no mid-run (\eta) change).

(a) (12 points). Which interval contains that median?

· 1. ([50, 150)) · 2. ([150, 250)) · 3. ([250, 350)) · 4. ([350, 450))

(b) (8 points). While riding near the edge, train loss over those late updates:

· 1. Strictly increases
· 2. Continues to decrease overall (Cohen-style)
· 3. Exactly flat machine-epsilon
· 4. Diverges to (\infty)

Q13. Ensemble vs endpoints on an early-fork cell

Difficulty: Medium · theme P1(a) · ledger `P1-ens-vs-endpoint`

From the assignment. Forked-run grid with `wsd0.0`; for each cell evaluate endpoints and (0.5/0.5) prediction ensemble (p_{\mathrm{ens}}).

This question changes. Restrict to cell ((x,y)=(0,500)), seeds `{1,2}`. Let (L_{\star} =\min(L(\theta_1),L(\theta_2))) and (G_{\mathrm{ens}}=L_{\star}-L(p_{\mathrm{ens}})).

(a) Sign (8 points). Which do you predict?

· 1. (G_{\mathrm{ens}}<0) (ensemble worse than best endpoint)
· 2. (G_{\mathrm{ens}}\in[0,0.01))
· 3. (G_{\mathrm{ens}}\ge 0.01) (ensemble clearly better than best endpoint)
· 4. Ensemble undefined (probabilities not comparable)

(b) Vs weight average (12 points). On the same cell, ranking of (L(p_{\mathrm{ens}})) vs (L(\theta_{\mathrm{avg}})):

· 1. (L(p_{\mathrm{ens}}) < L(\theta_{\mathrm{avg}})-0.01)
· 2. Within (0.01)

· 3. (L(p_{\mathrm{ens}}) > L(\theta_{\mathrm{avg}})+0.01)
· 4. (\theta_{\mathrm{avg}}) diverges

Q14. Extrapolate barrier growth along (x=0)

Difficulty: Medium · theme P1(b) · ledger `P1-B-x0-y750`
(Evidence: EXTRAPOLATED from Fit A — not MEASURED.)

From the assignment. P1 grid includes (y\in{25,50,100,250,500}) at each fork prefix (x).

This question changes. Hold (x=0). Extend post-fork horizon to (y=750) (beyond the measured grid). Use the Batch 1 growth pattern of (B) vs (y) at (x=0) (Appendix Q1.1 Fit A).

(a) (12 points). Which interval contains (B(0,750))?

· 1. ([0.5, 0.9)) · 2. ([0.9, 1.4)) · 3. ([1.4, 2.0)) · 4. ([2.0, \infty))

(b) (8 points). Relative to MEASURED (B(0,500)), you expect (B(0,750)):

· 1. Smaller (barrier shrinks with more branch steps)
· 2. About equal ((\lvert\Delta B\rvert<0.05))
· 3. Larger by (\gtrsim 0.3)
· 4. Negative (averaging becomes safe)

Q15. Mid-prefix fork at fixed long branch

Difficulty: Medium · theme P1(b) · ledger `P1-B-x25-y500`
(Evidence: EXTRAPOLATED from Fit B + MEASURED endpoints — not MEASURED at (x=25).)

From the assignment. Grid samples (x\in{0,50,100,250,500,1000}) at (y=500).

This question changes. Insert an unmeasured mid-prefix cell ((x,y)=(25,500)). Anchor on MEASURED (B(0,500)) and (B(50,500)) and the exponential (B)–vs–(x) fit at (y=500) (Fit B).

(a) (12 points). Which interval contains (B(25,500))?

· 1. ([0.10, 0.25)) · 2. ([0.25, 0.40)) · 3. ([0.40, 0.55)) · 4. ([0.55, \infty))

(b) (8 points). Ranking vs the two MEASURED anchors:

· 1. (B(25,500) < B(50,500) < B(0,500))
· 2. (B(50,500) < B(25,500) < B(0,500))
· 3. (B(25,500) > B(0,500))
· 4. (B(25,500)\le 0.01)

Q16. Longer shared prefix at (y=500)

Difficulty: Easy · theme P1(b) · ledger `P1-B-x2000-y500`
(Evidence: EXTRAPOLATED from Fit B asymptote — not MEASURED.)

From the assignment. Longest measured prefix is (x=1000); at (y=500), MEASURED (B(1000,500)<0).

This question changes. Extend shared prefix to (x=2000) at (y=500) (same `wsd0.0`, seeds).

(a) (12 points). Which interval contains (B(2000,500))?

· 1. ((-\infty, -0.12)) · 2. ([-0.12, -0.01)) · 3. ([-0.01, 0.01]) · 4. ((0.01, \infty))

(b) (8 points). Does averaging work ((B\le 0.01))?

· 1. Yes · 2. No · 3. Only if LR retuned · 4. Undefined

Q17. Ensemble–avg gap at extrapolated early fork

Difficulty: Hard · theme P1(a–b) · ledger `P1-G-x0-y750`
(Evidence: EXTRAPOLATED from Fit C ∘ Fit A — not MEASURED.)

From the assignment. On failure cells, (G=L(\theta_{\mathrm{avg}})-L(p_{\mathrm{ens}})) grows with (B).

This question changes. At the extrapolated cell ((0,750)), predict (G) using the Batch 1 (G)–vs–(B) fit composed with Fit A's (B(0,750)).

(a) (12 points). Which interval contains (G(0,750))?

· 1. ([0.5, 1.0)) · 2. ([1.0, 1.5)) · 3. ([1.5, 2.0)) · 4. ([2.0, \infty))

(b) (8 points). Relative to MEASURED (G(0,500)\approx 0.73):

· 1. (G(0,750)) smaller
· 2. About equal
· 3. Clearly larger ((\Delta G\gtrsim 0.3))
· 4. Gap sign flips ((G<0))

Q18. Medium branch, zero prefix

Difficulty: Easy · theme P1(b) · ledger `P1-B-x0-y250`

From the assignment. Same forked-run grid / barrier definition.

This question changes. Restrict to cell ((x,y)=(0,250)) (shorter branch than Q1's ((0,500))).

(a) (10 points). Which interval contains (B(0,250))?

· 1. ((-\infty, 0.01]) · 2. ((0.01, 0.05)) · 3. ([0.05, 0.2)) · 4. ([0.2, \infty))

(b) (10 points). Ranking vs MEASURED pattern at (x=0):

· 1. (B(0,250) < B(0,100) < B(0,500))
· 2. (B(0,100) < B(0,250) < B(0,500))
· 3. (B(0,500) < B(0,250))
· 4. All three (\le 0.01)

Q19. Borderline long-branch cell

Difficulty: Medium · theme P1(b) · ledger `P1-B-x100-y500`

From the assignment. Failure cells are those with (B>0.01).

This question changes. Restrict to ((x,y)=(100,500)).

(a) (10 points). Which interval contains (B(100,500))?

· 1. ((-\infty, 0.01]) · 2. ((0.01, 0.05)) · 3. ([0.05, 0.2)) · 4. ([0.2, \infty))

(b) (10 points). On this cell, median-style (G=L(\theta_{\mathrm{avg}})-L(p_{\mathrm{ens}})) is in:

· 1. ([0, 0.05)) · 2. ([0.05, 0.15)) · 3. ([0.15, 0.30)) · 4. ([0.30, \infty))

Q20. Grid-wide connectivity fraction

Difficulty: Easy · theme P1(a–b) · ledger `P1-grid`

From the assignment. Full (6\times 5) forked-run heatmap under `wsd0.0`.

This question changes. Report the fraction of the 30 measured cells with (B\le 0.01).

(a) (12 points). Which interval contains that fraction?

· 1. ([0, 0.2)) · 2. ([0.2, 0.5)) · 3. ([0.5, 0.8)) · 4. ([0.8, 1])

(b) (8 points). Where do the (B>0.01) failures concentrate?

· 1. Large (x), small (y)
· 2. Small (x), large (y)
· 3. Uniform across the grid
· 4. Only at (x=1000)

Q21. Fit-(C) gap from a mid-band barrier

Difficulty: Medium · theme P1(a–c) · ledger `P1-G-fitC-mid`
(Evidence: FITTED in-sample + MEASURED check on cell ((50,500)).)

From the assignment. Ensemble–avg gap (G) tracks barrier height when modes separate.

This question changes. Take MEASURED cell ((50,500)) with midpoint (B\in[0.05,0.2)). Using the linear (G)–vs–(B) relationship fitted on Batch 1 (Appendix Q1.1 Fit C), predict (G).

(a) (12 points). Which interval contains (G(50,500))?

· 1. ([0.10, 0.15)) · 2. ([0.15, 0.20)) · 3. ([0.20, 0.25)) · 4. ([0.25, 0.35))

(b) (8 points). Closest mechanistic reading of Fit C:

· 1. (G\approx B) (gap equals barrier)
· 2. (G\approx B+0.09) (gap ≈ barrier plus nearly constant ensemble win vs midpoint endpoints)

· 3. (G) independent of (B)
· 4. (G\approx -B)

Q22. Batch 2 schedule regret bands (jobs 1–4)

Difficulty: Medium · theme P2(a) · ledger `B2-P2-wsd0.0-R` **…** `B2-P2-wsd0.2-R` / `P2-sched-rank` **(Evidence: MEASURED — Batch 2 locked** `2026-10-11T00:32:07Z`**; finals from** `batch2_partial_finals.csv` **/ ledger Observed.)**

From the assignment. P2(a) four schedules at equal tokens; regret vs best raw final.

This question changes. Restrict to Batch 2 jobs `a3-b2-p2-wsd0.0`, `…-linear`, `…-cos`, `…-wsd0.2` (no dip). Let (R_M=L_M-L_{\mathrm{ref}}) with (L_{\mathrm{ref}}=\min) of the four finals.

(a) Constant schedule regret (10 points). Which interval contains (R_{\mathrm{wsd0.0}})?

· 1. ([0, 0.02)) · 2. ([0.02, 0.08)) · 3. ([0.08, 0.15)) · 4. ([0.15, \infty))

(b) Ranking (10 points). Closest final-val ranking:

· 1. (L_{\mathrm{wsd0.0}} < \min(L_{\mathrm{cos}},L_{\mathrm{wsd0.2}},L_{\mathrm{linear}}))
· 2. (\min(L_{\mathrm{cos}},L_{\mathrm{wsd0.2}},L_{\mathrm{linear}}) < L_{\mathrm{wsd0.0}}) and (\max(R_{\mathrm{cos}},R_{\mathrm{wsd0.2}},R_{\mathrm{linear}})<0.02)
· 3. All four within (0.005)
· 4. Linear uniquely best by (\ge 0.05)

Q23. Batch 2 horizon-free pair (jobs 6–7)

Difficulty: Hard · theme P2(d) · ledger `B2-P2-invsqrt-dL` / `B2-P2-sgdr-dL` **(Evidence: MEASURED invsqrt + SGDR (\Delta L) vs linear — both band MISS; |SGDR−invsqrt| HIT.)**

From the assignment. Invsqrt and SGDR with fixed ((w,T)); no budget-specific cooldown.

This question changes. Batch 2 jobs `a3-b2-p2-invsqrt` and `a3-b2-p2-sgdr2000` at the full 614M-token stop. Compare each to the best of `{cos, wsd0.2, linear}` from Q22. Let (\Delta L_M=L_M-L_{\mathrm{tuned}}).

(a) Invsqrt gap (10 points). Interval for (\Delta L_{\mathrm{invsqrt}}):

· 1. ((-\infty, -0.01)) · 2. ([-0.01, 0.01)) · 3. ([0.01, 0.08)) · 4. ([0.08, \infty))

(b) SGDR vs invsqrt (10 points). (\lvert L_{\mathrm{sgdr}}-L_{\mathrm{invsqrt}}\rvert) in:

· 1. ([0, 0.03)) · 2. ([0.03, 0.08)) · 3. ([0.08, 0.15)) · 4. ([0.15, \infty))

Q24. Batch 2 sharpness starters (jobs 8–9)

Difficulty: Easy · theme P3(a)i + P3(b) · ledger `B2-P3-sgd-lam` / `B2-P3-adamw-pre` **(Evidence: MEASURED — (a) median (\lambda_{\max}=245.1631473795680 7); (b) post-warmup median (\eta\lambda^{\mathrm{pre}}=30.987715914514).)**

From the assignment. Clean full-batch GD (N=32), (\eta=0.01), 1500 updates; minibatch AdamW with preconditioned sharpness.

This question changes. Only the two Batch 2 starter jobs: `a3-b2-p3-0-sgd` and `a3-b2-p3-1-adamw` (no mid-run (\eta) change).

(a) Full-batch median (\lambda_{\max}) (1200–1500) (10 points):

· 1. ([50, 150)) · 2. ([150, 250)) · 3. ([250, 350)) · 4. ([350, 450))

(b) Minibatch post-warmup median (\eta\lambda^{\mathrm{pre}}) (10 points):

· 1. ([0, 20)) · 2. ([20, 30)) · 3. ([30, 50)) · 4. ([50, \infty))

Q25. P5 exploratory: post-separation cooldown vs constant continue

Difficulty: Hard · theme P1×P2 · ledger `P5-B-cool-x0` / `P5-dB-cool` / `P5-B-const-ctrl` / `P5-rank-arms` / `P5-emergent-basin`
(Evidence: MEASURED — `P5-B-cool-x0` **MISS; `P5-dB-cool` **MISS**; `P5-B-const-ctrl` **HIT**; `P5-rank-arms` **MISS**; `P5-G-cool-fitC` **HIT**; `P5-emergent-basin` **MISS** (from `internal/p5-score-dryrun.md` / `media/p5/`))

From the assignment (reference only). P1 forked-run grid with `wsd0.0`. MEASURED cell ((x,y)=(0,500)) has midpoint barrier (B=0.6237>0.01) (weight averaging fails). P2 river-valley picture: large LR bounces on valley walls; decaying LR settles toward the river.

This question changes. Take the two ((0,500)) branch endpoints (seeds `{1,2}`). Continue **both** for (y_{\mathrm{cool}}=100) more updates **without** resetting model or optimizer state, under one of two arms, then re-measure (B):

- Arm C: learning rate decays to ~0 over those 100 updates (cosine or linear cooldown).
- Arm K: learning rate stays at the constant post-warmup value (`wsd0.0`).

Everything else (data distribution, seeds, eval protocol) matches P1. Distinct from Q4 / `P1-d-dB-cos` (cosine *during* the fork branch). Let (B_R=0.6237) (MEASURED raw midpoint).

(a) Barrier after cooldown (12 points). Which interval contains (B_C)? (Implies (\Delta B=B_R-B_C) in ledger `P5-dB-cool` **PREDICT** `[0.5,0.8)` if option 1.)

· 1. ((-\infty, 0.01]) · 2. ((0.01, 0.05)) · 3. ([0.05, 0.2)) · 4. ([0.2, \infty))

(b) Barrier after constant continue (8 points). Which interval contains (B_K)?

· 1. ((-\infty, 0.01]) · 2. ((0.01, 0.05)) · 3. ([0.05, 0.2)) · 4. ([0.5, \infty))

(c) Emergent interpretation (10 points). Which statement do you predict? (Also encodes ranking `P5-rank-arms`: (B_C < B_R) and (B_C < B_K).)

· 1. Same-valley walls: (B_C\le 0.01) and (B_K>0.01)
· 2. True disconnected modes: (B_C>0.05) and (B_K>0.05)
· 3. Cooldown irrelevant: (\lvert B_C-B_R\rvert<0.05)
· 4. Constant continue connects: (B_K\le 0.01) and (B_C>0.01)

Q26. Late-training sharpest Hessian family

Difficulty: Easy · theme P4(a) · ledger `P4-a-family-sharpest`
(Evidence: MEASURED @13/13 — late_sharpest=`head`** ((5.641603135007765)); ledger MISS vs PREDICT attention|mlp.)**

From the assignment. Measure provided AdamW checkpoints: per-family block (\lambda_{\max}) for `embedding`, `attention`, `qk_norm`, `mlp`, `norm`, `head`.

This question changes. Report which family is sharpest near the final step (9375 / `FINAL_STEP`), and how init differs.

(a) Late sharpest family (12 points). Closest prediction:

· 1. `embedding` uniquely sharpest
· 2. `head` uniquely sharpest
· 3. `attention` or `mlp` (tied-ok)
· 4. `qk_norm` or `norm`

(b) Init vs late (8 points). At step 0 vs 614M:

· 1. Same family always sharpest
· 2. `embedding`/`head` more often in top-2 at init than at late
· 3. Only `mlp` can be sharp early
· 4. No family structure at init

Q27. Spectrum / block energy over training

Difficulty: Medium · theme P4(a) · ledger `P4-a-density-neg` / `P4-a-block-energy`
(Evidence: dens-neg MEASURED MISS — neg_frac (0.00451\to\ldots\to0.06965) no shrink 0→9375; `P4-a-block-energy` **MEASURED HIT — excess_final=0.08727119221832025; fam_above=True; layer_above=True.)**

From the assignment. Density at init / 10M / 100M / 614M; split (\mathrm{tr}(H^2)) within vs between family blocks vs a random partition of equal sizes.

This question changes. Directional claims only (no invented magnitudes as MEASURED).

(a) Negatives (10 points). Closest spectrum story:

· 1. No negatives ever
· 2. Negatives at init; negative mass shrinks by 614M; final has clear positive outliers
· 3. Negatives grow to dominate at final
· 4. Density undefined on transformers

(b) Block energy (10 points). Within-family Frobenius fraction vs matched random partition at final:

· 1. Family (\le) random
· 2. Family (>) random; excess in ([0.05, 0.30))
· 3. Family (>) random; excess (\ge 0.50)
· 4. Only layer partition matters; family = random

Q28. Training recipe moves (\lambda_{\max}) inside the loss band

Difficulty: Hard · theme P4(b) · ledger `P4-b-train-2x`
(Evidence: PREDICTION only — P4 draft; freeze `TrainConfig` **before Batch 3** `b-train`**.)**

From the assignment. Move final (\lambda_{\max}) by (\ge 2\times) while keeping val in ([2.925,2.931]).

This question changes. One training-knob arm (WD / peak LR / embedding-init — not hand rescale). Baseline mid-band (\lambda\sim 5.25).

(a) (12 points). Joint outcome you predict:

· 1. (\lambda) factor (\ge 2\times) **and** val in ([2.925,2.931])
· 2. (\lambda) factor (\ge 2\times) but val exits the band (still "success")
· 3. Val stays in band but (\lambda) moves (<2\times)
· 4. Both criteria fail

(b) (8 points). If val exits the band:

· 1. Widen the band and count success
· 2. Count as **failed arm** (do not relabel)
· 3. Drop (\lambda) criterion
· 4. Use train loss instead of val

Q29. Subspace (k) ablation

Difficulty: Medium · theme P4(c) · ledger `P4-c-k-ablate`
(Evidence: PREDICTION only — optional Batch 3 follow-on; not measured.)

From the assignment. Default `k=10`, refresh 25, 300 steps from SGDM @ 1526.

This question changes. Repeat `top` / `removed` at (k=3) and (k=30) (same start/refresh/steps).

(a) Mean grad fraction at (k=3) (10 points):

· 1. ([0.15, 0.35)) · 2. ([0.35, 0.65)) · 3. ([0.65, 0.85)) · 4. ([0.85, 1])

(b) At (k=30) vs (k=10) (10 points):

· 1. Fraction lower; removed–full gap larger
· 2. Fraction higher (([0.75,0.95))); top even closer to full
· 3. Identical to (k=10)
· 4. Removed becomes best arm

Answer sheet (practice)

Name _____ SUNet ID _____

Q	Part	Your option	Assignment problem cited	Observations (MEASURED / FITTED only)	Rationale
Q1	a		P1(b)		
Q1	b		P1(b)		
Q2	a		P1(a–b)		
Q2	b		P1(a–b)		
Q3	a		P1(c)		
Q3	b		P1(c)		
Q4	a		P1(d)		
Q4	b		P1(d)		
Q5	a		P2(b)		
Q5	b		P2(b)		
Q6	a		P2(c)		
Q6	b		P2(c)		
Q7	a		P3(a)ii		
Q7	b		P3(a)ii		
Q8	a		P3(b)		
Q8	b		P3(b)		
Q9	a		P4(b)		
Q9	b		P4(b)		
Q10	a		P4(c)		
Q10	b		P4(c)		
Q11	a		P2(d)		
Q11	b		P2(d)		
Q12	a		P3(a)i		
Q12	b		P3(a)i		
Q13	a		P1(a)		
Q13	b		P1(a–b)		
Q14	a		P1(b)		
Q14	b		P1(b)		
Q15	a		P1(b)		
Q15	b		P1(b)		
Q16	a		P1(b)		
Q16	b		P1(b)		
Q17	a		P1(a–b)		
Q17	b		P1(a–b)		
Q18	a		P1(b)		
Q18	b		P1(b)		
Q19	a		P1(b)		
Q19	b		P1(a–b)		
Q20	a		P1(a–b)		
Q20	b		P1(b)		
Q21	a		P1(a–c)		
Q21	b		P1(a–c)		

Q	Part	Your option	Assignment problem cited	Observations (MEASURED / FITTED only)	Rationale
Q22	a		P2(a) B2		
Q22	b		P2(a) B2		
Q23	a		P2(d) B2		
Q23	b		P2(d) B2		
Q24	a		P3(a)i B2		
Q24	b		P3(b) B2		
Q25	a		P5 expl.		
Q25	b		P5 expl.		
Q25	c		P5 expl.		
Q26	a		P4(a)		
Q26	b		P4(a)		
Q27	a		P4(a)		
Q27	b		P4(a)		
Q28	a		P4(b)		
Q28	b		P4(b)		
Q29	a		P4(c)		
Q29	b		P4(c)		

Answer key

How to read. Keep the original **PREDICTION** option. Fill **Resolved** only when evidence exists (**MEASURED**, **FITTED**, or **EXTRAPOLATED**). Never erase a wrong PREDICTION — the miss is the learning signal. Leave Resolved = **—** with tag **PENDING** when the metric is missing (mid-dip, EMA, P3 η-change, mid-budget HF, P4 optional train/k-ablate, `P5-L-cool-dir`). **P5** otherwise MEASURED 6/7 (see Q25 / rule 8). P4 stage-a **13/13** ledger apply: Q26 / Q27a from ledger HIT/MISS; Q27b Resolved **HIT** (MEASURED excess_final=0.08727119221832025). Do **not** leave SGDR / P3 clean-edge / AdamW precond blank — those are MEASURED @ Batch 2 9/9.

Sources for MEASURED cells: `media/p1/barrier-table.csv`, `media/p1/barrier-summary.json`, `media/p1/a3-mode-connectivity.json`, `media/p2/batch2_partial_finals.csv` (jobs 0–6), `media/p2/p3_sgd_sharpness_summary.json`, `media/p2/p3_adamw_sharpness_summary.json`, P4 13/13: `media/p4/stage-a/a-adamw-*.json` ((\lambda@9375=5.967310256347902); ledger `docs/a3-prediction-ledger.md`). Fits: intuition guide Appendix Q1.1. Batch 2 PREDICTIONs locked `2026-10-11T00:32:07Z` — mid-run W&B not used.

Q	Part	PREDICTION (option)	Resolved	Tag	Mechanism / evidence
Q1	a	**3** `[0.05, 0.2)`	**4** `[0.2, ∞)` · (B=0.6237)	MEASURED	Early fork + long branch → large barrier (mag
Q1	b	**1** `(−∞, 0.01]`	**1** · (B=-0.0597)	MEASURED	Long prefix + short branch → averaging work
Q2	a	**3**	**3** · all four failure (G\ge 0.01)	MEASURED	Output ensemble bypasses parameter-space r
Q2	b	**2** `[0.02, 0.08)`	**4** `[0.2, ∞)` · median (G\approx 0.279)	MEASURED	Gaps (0.1311,0.2309,0.3273,0.7281) (magnitu
Q3	a	**1**	**1** · (\alpha=0.5) worst on ((50,500))	MEASURED	Unimodal midpoint barrier when (B) large (hi
Q3	b	**3** `[0.05, 0.2)`	**3** · (B=0.1299)	MEASURED	Mid-band cell ((50,500)) (hit)
Q4	a	**3** `[0.02, 0.1)`	—	PENDING	`P1-d-dB-cos` · no cosine-fork MEASURED
Q4	b	**3** Depends on (B_W)	—	PENDING	Note: MEASURED (B_W(250,250)=-0.0779) under `wsd0.0`
Q5	a	**3** `[0.5, 0.75)`	—	PENDING	B2-locked · `P2-ema-gap-frac` · no EMA gap M
Q5	b	**3**	—	PENDING	B2-locked · no EMA ranking MEASURED
Q6	a	**2** `[−0.02, 0.02)`	**2** · (R=0.0012464523315429688) vs `wsd0.2`	MEASURED	`rms7iyyt−soosjcxr`; band HIT
Q6	b	**1** `(−∞, −0.02)`	—	PENDING	mid-dip not in `batch2_partial_finals.csv`
Q7	a	**3** `[350, 450)`	—	PENDING	η-half arm not run · PREDICTION (2/0.005=
Q7	b	**2** `[80, 120)`	—	PENDING	η-double arm not run · PREDICTION (2/0.02
Q8	a	**3** `[30, 50)`	**3** · (\eta\lambda^{\mathrm{pre}} =30.987715914514)	MEASURED	`B2-P3-adamw-pre` · HIT near 38
Q8	b	**2**	**2** · plateau (\approx 30.15); not stuck at 2	MEASURED	noisier band near ~38 vs full-batch (\eta\lam
Q9	a	**1** `[0, 0.002)`	**4** ·	ΔL	=2.73853462934494
Q9	b	**4** or **1**	**1** · ratio=0.4658784038724411	MEASURED	`P4-invar-dL` · **MISS** · file-exact from dry-run
Q10	a	**3** `[0.6, 0.85)`	**1** · mean=0.09296940652964016	MEASURED	`P4-top10-frac` · **MISS** · file-exact from dry-ru
Q10	b	**1**	**3** · gap=-0.000344693660736084	MEASURED	`P4-c-loss-rank` · **MISS** · L=(3.582527816295624,5.502330541610718,
Q11	a	**3** `[0.01, 0.08)`	**4** · (\Delta L=0.09225320816040039\in[0.08,\infty))	MEASURED	`1jri34em` vs linear; band MISS (SGDR likewis
Q11	b	**2**	—	PENDING	Mid-budget stop not in Batch 2 required stop
Q12	a	**2** `[150, 250)`	**2** · median (\lambda_{\max} =245.16314737956807)	MEASURED	`B2-P3-sgd-lam` · `0dewkykj` / `p3_sgd_sharpness`
Q12	b	**2**	**2** · late train ↓ (mid≈2.14@1224 → final 1.2686@1499)	MEASURED	Cohen-style continue-decrease; part of `B2-P3-`
Q13	a	**2** `[0, 0.01)`	**3** · (G_{\mathrm{ens}} =3.7483-3.6454=0.1029)	MEASURED	Soft ensemble beats best endpoint by (\ge 0.0 miss vs PREDICTION)
Q13	b	**1**	**1** · (L_{\mathrm{ens}}=3.6454) vs (L_{\mathrm{avg}}=4.3734)	MEASURED	High-(B) cell; ensemble ≪ weight avg (hit)
Q14	a	**2** `[0.9, 1.4)`	**2** · growth-lin (\approx 1.03); quad (\approx 1.22)	EXTRAPOLATED	Fit A beyond (y=500); not MEASURED
Q14	b	**3**	**3**	EXTRAPOLATED	Growth regime: longer (y) raises (B) at (x=0)
Q15	a	**2** `[0.25, 0.40)`	**2** · exp (\approx 0.319); lin-interp (\approx 0.377)	EXTRAPOLATED	Fit B + MEASURED ((0,500),(50,500))
Q15	b	**2**	**2**	EXTRAPOLATED	Monotone in (x) between anchors
Q16	a	**2** `[-0.12, -0.01)`	**2** · asymptote (\approx -0.066)	EXTRAPOLATED	Fit B (x\to\infty) at (y=500)
Q16	b	**1** Yes	**1**	EXTRAPOLATED	Asymptote well below (0.01)

Q	Part	PREDICTION (option)	Resolved	Tag	Mechanism / evidence
Q17	a	**2** `[1.0, 1.5)`	**2** · (G(B_{\mathrm{lin}})\approx 1.15); (G(B_{\mathrm{quad}})\approx 1.35)	EXTRAPOLATED	Fit C ◦ Fit A
Q17	b	**3**	**3**	EXTRAPOLATED	Larger (B) ⇒ larger (G)
Q18	a	**4** `[0.2, \infty)`	**4** · (B=0.2173)	MEASURED	Failure cell ((0,250))
Q18	b	**2**	**2** · (B(0,100)=-0.0275 < B(0,250)=0.2173 < B(0,500)=0.6237)	MEASURED	Growth along (x=0)
Q19	a	**2** `(0.01, 0.05)`	**2** · (B=0.0319)	MEASURED	Mildest failure cell
Q19	b	**2** `[0.05, 0.15)`	**2** · (G=0.1311)	MEASURED	Smallest failure gap
Q20	a	**2** `[0.2, 0.5)`	**4** `[0.8, 1]` · (26/30=0.867)	MEASURED	Connectivity under-predicted
Q20	b	**2**	**2** · failures ((0,250),(0,500),(50,500),(100,500))	MEASURED	Small (x), large (y)
Q21	a	**3** `[0.20, 0.25)`	**3** · FITTED (G\approx 0.233); MEASURED (G=0.2309)	FITTED+MEASURED	Fit C on mid-band cell (hit)
Q21	b	**2**	**2** · MEASURED (\bar L-L_{\mathrm{ens}}) mean (\approx 0.089)	FITTED	(G\approx B+0.09)
Q22	a	**2** `[0.02, 0.08)`	**4** `[0.15, \infty)` · (R=0.16930222511291504) (band miss; worst-of-four hit)	MEASURED	`B2-P2-wsd0.0-R` · constant worse than band
Q22	b	**2**	**2** · linear < cos < wsd0.2≪wsd0.0; (R_{\mathrm{lin}}=0) (R_{\mathrm{cos}} =0.005916595458984375) (R_{\mathrm{wsd0.2}} =0.009538650512695312); (L): linear(2.9282028675079346) < cos(2.934119462966919) < wsd0.2(2.93774151802063) < wsd0.0(3.0975050926208496) (rank HIT; wsd0.2 top-2 MISS)	MEASURED	`P2-sched-rank` + `B2-P2-linear-R` / `B2-P2-cos`
Q23	a	**3** `[0.01, 0.08)`	**4** · (\Delta L=0.09225320816040039\in[0.08,\infty))	MEASURED	`B2-P2-invsqrt-dL` · `1jri34em` vs linear; band
Q23	b	**1** `[0, 0.03)`	**1** · (\lvert L_{\mathrm{sgdr}}-L_{\mathrm{invsqrt}} \rvert=0.014348745346069336); (\Delta L_{\mathrm{sgdr}} =0.10660195350646973) band MISS vs linear	MEASURED	`B2-P2-sgdr-dL` · job 6 `51b3qq4n` @ 9/9 · (L_{\ =3.0348048210144043) (L_{\mathrm{linear =2.9282028675079346)
Q24	a	**2** `[150, 250)`	**2** · median (\lambda_{\max} =245.16314737956807)	MEASURED	`B2-P3-sgd-lam` · HIT near (2/\eta=200)
Q24	b	**3** `[30, 50)`	**3** · (\eta\lambda^{\mathrm{pre}} =30.987715914514)	MEASURED	`B2-P3-adamw-pre` · HIT
Q25	a	**1** `(-∞, 0.01]`	**4** · B_C=0.7697049379348755; dB=-0.14600493793487546 (B_R=0.6237 [JSON] − B_C=0.7697049379348755)	MEASURED	`P5-B-cool-x0` · **MISS** · file-exact from dry-run
Q25	b	**4** `[0.5, ∞)`	**4** · B_K=0.7320129871368408	MEASURED	`P5-B-const-ctrl` · **HIT** · file-exact from dry-r
Q25	c	**1** same-valley walls	**2** · B_C=0.7697049379348755 B_R=0.6237 B_K=0.7320129871368408	MEASURED	`P5-emergent-basin` · **MISS**; `P5-rank-arms` · **M** dry-run
Q26	a	**3** attention or mlp	**2** · late=`head` ((5.641603135007765))	MEASURED	`P4-a-family-sharpest` · **MISS** (not attention (7.38\notin[1.5,5))

Q	Part	PREDICTION (option)	Resolved	Tag	Mechanism / evidence
Q26	b	**2**	**2** partial · early_top2=`embedding,attention`; late=`head`	MEASURED	init has embed (not head); late head-led — ro ranking
Q27	a	**2**	shrink failed · neg_frac (0.00451\to0.01491\to0.93446\to0.06965)	MEASURED	`P4-a-density-neg` · shrink_0→9375=false · **M** (\in[0,0.15)))
Q27	b	**2** excess `[0.05,0.30)`	**2** · excess_final=0.08727119221832025; fam_above=True; layer_above=True	MEASURED	`P4-a-block-energy` · **HIT** · file-exact from dry
Q28	a	**1**	—	PENDING	`P4-b-train-2x` · b not started
Q28	b	**2** failed arm	—	PENDING	`P4-b-train-2x` · b not started
Q29	a	**2** `[0.35, 0.65)`	—	PENDING	`P4-c-k-ablate` · c not started
Q29	b	**2**	—	PENDING	`P4-c-k-ablate` · c not started

How to promote PREDICTION → MEASURED / confirm EXTRAPOLATED

1. Run the corresponding starter / ledger ID (or measure an extrapolated cell).
2. Write the measured option in **Resolved** with tag **MEASURED**.
3. If MEASURED ≠ PREDICTION, keep both; update intuition guide §A/D and mark ledger row `resolved`.
4. For EXTRAPOLATED rows: leave Observed empty until a future measure; do not paste fit points into Observed as if measured.
5. Never erase a wrong PREDICTION without recording the miss — that is the learning signal for the Oct 20 quiz.
6. For Batch 2 (`B2-*` / Q5–Q6, Q8, Q11–Q12, Q22–Q24): promote only from **final** metrics — never from mid-run W&B snapshots. All nine `B2-*` P2/P3 band rows promoted @ 9/9 (incl. both P3 sharpness HITs); Q5 EMA / Q6b mid-dip still PENDING.
7. For P4 @ **13/13**: Q26 / Q27a filled from ledger apply (`family` **MISS**, dens-neg **MISS**; sibling `norm-sharpness` **HIT**, `lam-9375-earlyfit` **MISS**). Q27b Resolved **HIT** from ledger apply (excess_final=0.08727119221832025). b/c scored: `P4-invar-dL` **MISS**, `P4-top10-frac` **MISS**, `P4-c-loss-rank` **MISS**. Q28–Q29 stay PENDING without train/k-ablate. Status: `internal/p4-bc-apply-readme.md` · dry-runs under `internal/`.
8. For P5 exploratory (Q25 / `P5-*`): scored MEASURED: `P5-B-cool-x0` **MISS**, `P5-dB-cool` **MISS**, `P5-B-const-ctrl` **HIT**, `P5-rank-arms` **MISS**, `P5-G-cool-fitC` **HIT**, `P5-emergent-basin` **MISS**; `P5-L-cool-dir` **INSUFFICIENT** (`dL=null` / (L_R,L_C) missing — no HIT/MISS). Keep PREDICTION options if miss; see `internal/p5-fill-on-complete.md`.